AI and Ethics
https://doi.org/10.1007/s43681-023-00272-x

ORIGINAL RESEARCH

The ethics of computer vision: an overview in terms of power

Rosalie A. Waelen[1]

Received: 4 January 2023 / Accepted: 27 February 2023
© The Author(s) 2023

Abstract
Computer vision is a subfield of artificial intelligence, aimed at making computers see. Computer vision tools enable a system or device to automatically analyze, interpret, and respond to images and videos. Computer vision tasks range from object detection and tracking, to the recognition of people's faces and emotional states. While the ethics of AI in general has received significant attention, and the ethics of facial recognition (a computer vision application) too, little of the AI ethics literature focuses specifically on the ethics of computer vision. In this chapter, I create an overview of ethical, social, and political issues related to computer vision, using a critical approach. This means that I identify issues in terms of power and evaluate them in function of their impact on the value of autonomy and the normative goal of emancipatory progress. The aim of this chapter is first and foremost to offer an overview of potential normative implications of computer vision. Additionally, the chapter functions as an example for the use of a critical approach to AI ethics.

Keywords Computer vision · Video analytics · Facial recognition · AI ethics · Power

1 Introduction

Computer vision (CV) involves the use of artificial intelligence (AI) techniques to automate the analysis of images and videos. CV comprises several tasks, including tracking, identification, detection, classification, localization, segmentation, facial recognition, emotion recognition, and behavior recognition. These tasks can serve a wide variety of purposes: from reading handwritten texts to recognizing traffic signs, and from interpreting MRI-scans to sentiment analysis of audiences. Given the many possible use cases of computer vision, it is easy to imagine that the technology can significantly impact everyday life and societal practices. This article provides an overview of the potential ethical, social, and political implications of CV. The overview focuses on those CV applications that analyze the visual data of human persons. The method used to create the overview is a critical approach, based on a pluralistic understanding of power (as proposed by Waelen [25]).

The outline of the article is as follows. In Sect. 2, I discuss related work on the ethics of CV. In Sect. 3, I introduce the critical approach used to analyze CV's potential normative implications. In Sects. 4, 5, 6 and 7, I apply the approach to identify and evaluate the issues in CV related to dispositional power, episodic power, systemic power, and constitutive power. I end the chapter with a conclusion in Sect. 8.

2 Related work

A quick narrative review of the AI ethics literature shows that there is little work on the ethics of CV, despite the widespread attention for AI ethics and the importance of CV within AI research.

For starters, in general discussions of AI ethics, CV and its different applications are hardly mentioned.[1] The *Stanford Encyclopedia of Philosophy* entry on 'Ethics of Artificial Intelligence and Robotics' [20] and the *Internet Encyclopedia of Philosophy* entry on 'Ethics of Artificial Intelligence' [11] both only refer to facial recognition once and do not explicitly mention CV at all. In *The Oxford Handbook of Ethics of AI* [9], there is not a single chapter dedicated to CV or CV applications—although CV applications are among the examples discussed in some of the chapters. In Mark Coeckelbergh's book *AI Ethics* [4], computer vision is briefly mentioned as one of many AI techniques and facial

✉ Rosalie A. Waelen
r.a.waelen@utwente.nl

1 Section of Philosophy, University of Twente, Enschede, The Netherlands

[1] The same could be said for Natural Language Processing (NLP), which is another main subfield of AI.

recognition repeatedly pops up in examples (regarding surveillance and privacy, data and biases, Walmart's analysis of customers, and Facebook's photo tagging), but neither the technology nor the examples are discussed with much detail. In Kate Crawford's book *The Atlas of AI. Power, Politics, and the Planetary Cost of Artificial Intelligence* one can find more detailed discussions of CV—although still only in relation to examples that are meant to portray other issues. The discussion of CV in Crawford's book shows a lot of overlap with her essay on the normative issues in CV research that was published on the webpage *excavating.ai* [7]. For instance, she discusses the enormous amount of energy needed to store the big datasets needed to develop CV models, the problematic assumption that images are apolitical and can be given a single label, and the problems with Ekman's theory of the facial expressions of emotions on which most CV-based emotion recognition models build.

However, despite the lack of attention for CV in general works on AI ethics, there are some focused articles dealing explicitly and exclusively with the ethics of CV or certain CV applications. Already in 2004, Brey discussed the ethics of the use of facial recognition in public spaces. He highlights the problem of error, the problem of function creep, and privacy—noting that the latter is the most serious obstacle to facial recognition applications in public space [2]. Selinger and Leong [22] wrote a chapter on the ethics of facial recognition in *The Oxford Handbook of Digital Ethics*, discussing among other things the uniqueness of facial recognition technology. In their much-cited paper 'Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification' [3], Buolamwini and Gebru discuss algorithmic fairness in the context of facial analysis systems that classify a person's gender based on the image of their face. Thiago Guimarães Moraes and colleagues discuss different purposes in which facial recognition is used in (semi-)public spaces in Brazil (namely for public security, social scoring and digital identity, private security, targeted marketing, and public health) and the risks that come with these applications (the lack of a legal basis, inaccuracy, normalization of surveillance, and a lack of transparency) [18]. Blank et al. analyze the ethics of facial recognition in general. Although they claim to build on various approaches in AI ethics, such as the expansion of bioethics into AI suggested by the AI4People framework, they end up focusing on three issues—human rights, error rates, and bias—without explaining why these would be the most important to discuss [1].

There are also a number of focused articles on aspects of CV other than face recognition or facial analysis. Crawford's aforementioned piece on *excavating.ai* offers an in-depth discussion of the problems surrounding ImageNet—the main database used in CV research. Coupland et al. discuss how ethical consideration could or should be part of the development of CV applications, on the basis of a specific case study—namely, a CV system for person tracking, occupancy and fall detection [5]. Huffer et al. go into the ethics of CV in the context of human remains trafficking [14]. Dufresne-Camaro et al. wrote a paper on CV research for the global South and the risks related to the uses of CV in the global South. They find that the focus of CV research is different in the global South and the risks related to CV applications and uses are "region-specific, depending heavily on a community's needs, norms, culture and resources" [10, p. 8]. In a report for the American Civil Liberties Union, Stanley [24] discusses the dangers of AI cameras and video analytics. Although Stanley's report focuses solely on the use of CV for surveillance, it covers a fairly wide range of ethical issues—including chilling effects, the new types of data smart cameras can gather, the unscientific basis of certain forms of analytics, discriminatory effects, and the possibility of over-enforcement and abuse of the technology.

To the best of my knowledge, only two sources cover the ethics of CV in general, rather than focusing on a specific application or use context. The first source is a Master's thesis, in which six ethical themes in computer vision are identified on the basis of a literature review: espionage, identity theft, malicious attacks, copyright infringement, discrimination, and misinformation [15]. The second piece on the ethics of CV is a conference paper by Skirpan and Yeh, titled 'Designing a Moral Compass for the Future of Computer Vision using Speculative Analysis'. The authors categorize five risks in CV: privacy violations, discrimination, security breaches, spoofing and adversarial inputs, and psychological harms [23].

Computer vision research covers a huge part of AI research and the developments in AI over the past decade started with a breakthrough in CV research [17]. When it comes to AI ethics, however, CV is much less central. Admittedly, the topic is not entirely ignored, and facial recognition has even become one of the most discussed topics in AI ethics. But on the basis of this brief literature review, I conclude that a comprehensive overview of potential ethical, social, and political issues in CV would nevertheless be a valuable addition to the AI ethics literature.

3 Method

To create an overview of CV's ethical, social, and political implications (CV's normative implications, for short), I opt for a critical approach (as proposed by Waelen [25]). The critical approach is inspired by the tradition of critical theory, which focuses on critiquing power dynamics and has the practical aim of realizing emancipatory progress. The critical approach has multiple benefits: it covers not only strictly ethical issues, but also social and

Table 1 Dispositional power

How can CV disempower people?	How could this affect autonomy and emancipatory progress?
Less informational control	Challenges ability to own one's life story
Less anonymity	Challenges ability to act and develop independently
Less epistemic agency due to technical and legal complexities surrounding CV practices	Challenges ability to follow one's own judgment and potentially harms self-development
Less testimonial agency due to automated interpretation of identity	Challenges ability to own one's life story
Less acting out of moral duty due to possibility of over-enforcement using CV	Challenges ability to follow one's own moral judgment and develop one's own moral reasoning
Loss of important skills due to dependency on CV applications	Challenges ability to follow one's own judgment

political implications; it avoids abstract ethical terminology; and it has a clear normative goal, namely furthering emancipatory progress. Ethical, social, and political implications of CV are identified and evaluated using a power framework. The power framework is based on a pluralistic understanding of the concept of power [13, 21, 25].

Haugaard [13] has argued that one should treat power as a family resemblance concept. This entails that different definitions of power that are usually taken to be opposing views, are treated as different aspects of the same thing and therefore as complementary views of power. Each aspect of power that is included in the framework, is included because it is useful for understanding the ways in which AI applications like CV might threaten or compromise human autonomy and emancipatory progress. The four aspects of power that make up the framework for the analysis are (1) *dispositional power*, or the potential to bring about significant outcomes; (2) *episodic power*, or the exercise of power by A over B, which makes B act differently than they otherwise would have; (3) *systemic power*, or the dominant laws, norms and practices in a society; and (4) *constitutive power*, or the ways in which power shapes a person's identity, thoughts and behavior [13, 21].

In what follows I devote a separate section to each of these four aspects of power. In each section I identify potential normative issues related to the respective form of power. I evaluate the identified issues by discussing their impact on autonomy and, consequently, emancipatory progress in society. For the purposes of this paper, I understand autonomy as the ability to follow one's own judgment (self-legislation) and the ability to develop and own one's life story (self-ownership). People form their own judgment and their life story in relation to and communication with others. The focus of the chapter lies predominantly on applications that analyze images or videos of persons. Furthermore, it should be noted that CV can also impact people and emancipation in a positive way. However, the present analysis focuses only on the potentially problematic impact of CV.

4 Dispositional power

Let us start the analysis with the dispositional view of power. This view, quite obviously, holds that power is a disposition or ability. However, one should not think of every human disposition or ability in terms of power. Dispositional power is the ability to bring about significant outcomes [19]. People are empowered when they gain dispositional power and disempowered when they lose it. When an individual or group is empowered or disempowered, their scope of agency increases or decreases. More agency promotes autonomy, it namely means that people are better able to act as they see fit, to develop themselves, and to explore opportunities. So, empowerment is a good worth striving for and disempowerment an ethical problem. When considering ethical and other normative issues in CV, it is therefore worthwhile asking: How can CV empower or disempower individuals? And: Do CV's disempowering effects limit people's autonomy? In this section, I identify six ways in which CV applications could potentially disempower people and discuss how these forms of disempowerment relate to autonomy (as defined above). The findings are summarized in Table 1 at the end of the section.

First of all, like many other modern technologies, CV limits people's informational control. CV applications retrieve information about individuals from visual data that is either pre-existing footage or gathered in real-time (e.g. through a camera on a personal device or a CCTV camera in a public place). When dealing with stored, pre-existing visual data, the persons displayed on the image or video (as well as the people taking the footage) most likely did not know that the footage would later be used as input for CV applications. In the case of real-time recordings, it can be difficult for a person to avoid being

recorded, because that would mean they could not freely move in public places or could not use certain devices or applications. So, in both cases, CV seriously challenges people's ability to consent to their data being gathered and analyzed. On top of that comes the fact that it is particularly difficult to hide or protect one's data from CV tools, because it is people's appearance that is revealing information in this case. For example, video analytics systems analyze a person's facial expressions or the way they walk. Protecting one's information, in this context, would imply taking extensive measures, such as hiding one's face or changing the way one dresses.

These threats to informational control can be understood as disempowering. Less informational control harms autonomy, because it limits an important ability: the ability of a person to decide who knows what about them. This ability is an element of self-ownership. Understanding informational control in terms of disempowerment also enables us to see how the concern for informational control, being a concern for privacy, derives from a more fundamental concern for human emancipation. People do not value informational control for its own sake, but because it promotes their autonomy.

Secondly, related to the loss of informational control is a diminishing ability to go places anonymously. Anonymity is not just something that benefits criminals or people with bad intentions, anonymity helps people to "feel freer to associate with whomever they want, read and watch what they choose, and express their opinions as they see fit" ([8], p. 210). In other words, anonymity creates the safe space that people sometimes need to act as they wish to and develop themselves. For example, public anonymity allows a person to visit a LGBTQ bar or attend a political protest, without their friends, family, or colleagues being able to find out. Anonymity also safeguards those fleeing oppressive regimes or those who, for good reasons, had to take on a new identity. So being able to go places anonymously is empowering in that it promotes the ability to act and develop freely and independently. A threat to anonymity is therefore also a threat to autonomy.

Thirdly, CV can decrease people's epistemic agency. Not only does CV challenge the extent to which people control what data is gathered and what is known about them, advanced data analysis also makes it increasingly difficult for the average individual to grasp what information is or could be retrieved from visual data. A person might therefore not know what information may be revealed when they are filmed by a 'smart camera' (as CV supported cameras are often called) or when they share images of themselves online. Furthermore, in addition to the technical complexity, there is also complexity regarding what data can legally be gathered and shared with third parties. This additional layer of complexity, or illiteracy, contributes to the loss of epistemic agency that CV applications can cause. A loss of epistemic agency due to the technical and legal complexity of CV applications relates to dispositional power in that it implies a decline in people's ability to follow their own judgment. Since I understood self-legislation as an important element of autonomy, I conclude that CV's effect on epistemic agency can threaten autonomy. Furthermore, the feeling of lacking sufficient epistemic agency to navigate the technical world around oneself, might also impact a person's self-confidence. Such an impact can also be understood as limiting autonomy, harming self-development namely implies harming a person's identity development.

Fourth, people lose testimonial agency when CV systems deprive them of the opportunity to communicate who they are and how they feel. Some CV applications automatically infer people's identity, demographic information, mood, or other characteristics. As a result, people are no longer able to communicate this information themselves. The ability to communicate one's own identity is important, not only because it allows people to control how much is known about them, but also because it enables them to shape their identity and exercise control over how they are perceived and treated by the world around them. In other words, testimonial agency supports the ability to own one's life story and thus supports autonomy. The way people present themselves is not objective: they stress the characteristics that they find most important and leave out those that they do not want others to know about of or associate them with. Maybe a person strongly identifies with a specific characteristic, such as their religion or ethnicity. Or conversely, maybe a person prefers to hide a certain characteristic, because they want to avoid being treated in discriminatory or stereotypical ways. CV tools could infer objective facts about people, such as their age and gender, but they cannot read from one's face what their subjective sense of identity is or who they aspire to be.

Fifth, CV might also affect people's ability to act out of moral duty. Hale [12] develops this idea in his 'subjective freedom argument' against face recognition technology. Hale argues that facial recognition will enable law-enforcement to become (close to) complete—meaning that every deviation from the law, every small wrongdoing (think jaywalking or littering), can be detected and penalized. Hale explains that such over-enforcement would be a moral problem, because moral behavior would no longer stem from people's own moral reasoning or inner moral laws, but only be the result of people's awareness that they are being watched and will be penalized if they do not act according to society's laws. Without a need to act out of duty, following one's own moral reasoning, citizens do not get the opportunity to develop their moral autonomy. In other words, CV could diminish people's ability to follow as well as people's ability to develop their own moral judgment. Of course traditional video surveillance could affect moral autonomy too, but

CV automated surveillance would make law-enforcement that much more effective that the impact on moral autonomy could be more significant too.

Finally, there are risks related to depending too much on a technology. The use of CV applications can result in deskilling, i.e., the loss of important human skills. This would for example be the case when people learn to depend on facial recognition systems to identify people or read their emotions, and thereby forget how to remember names and faces and interpret facial expressions. Or when people depend solely on photo translator applications when finding their way in a foreign country. Dependency or reliance on technology can be disempowering. Such disempowerment is problematic to the extent that it prevents a person from following their own judgment (self-legislation) or forming their own life story (self-ownership). Particularly in the example of deskilling one could argue that self-legislation is at stake.

5 Episodic power

The term 'episodic power' refers to the exercise of power by one actor over another. This view of power is therefore also often referred to as 'power-over'. One has power over another when they get them to act or think in a way they otherwise would not have acted or thought. Technology can facilitate the exercise of power "by either giving them new powers or by improving the efficiency, effectiveness, reliability and ease by which existing powers are exercised [2, p. 81]. So CV applications can function as tools that enable the exercise of power, but they can also simply add force to power that would have been exercised anyway. This is not necessarily bad news. Power relations are not always problematic they are an unavoidable part of life (think of parents having power over their children, or employers having power over their employees). In line with critical theory's emancipatory aim, we can say that episodic power becomes a moral concern when an exercise of power significantly compromises a person's self-legislation and self-ownership. Hence, the impact of a power relation on autonomy needs to be proportional to be morally acceptable.

A big part of CV's use for the exercise of power lies in its potential to automate surveillance. Surveillance makes it possible to control the way people act, either by detecting or punishing it when people act wrongly, or because people act differently when they are aware of being monitored. Before, camera surveillance depended on human operators to monitor video footage. Now that the monitoring of video footage is automated by CV, camera surveillance can be implemented on a much wider scale. As a consequence, surveillance also emerges in new contexts—take for example parents surveilling their children, health providers surveilling elderly or sick people in their homes, or insurance companies surveilling drivers in their cars. CV's automation of surveillance therefore facilitates the exercise of power by many parties: by law enforcement and security, by insurance companies, by employers, by private persons, and more.

As already pointed out in the previous section, CV enabled surveillance could lead to over-enforcement. By automatically detecting every single instance of misbehavior, however small, law enforcement could become too restrictive. While it is still enforcement of the law, which in itself is not problematic according to most people, over-enforcement could make the power law has over citizens too restrictive of people's freedom to act as they see fit (hence, their self-legislation).

CV enabled surveillance by insurance companies strengthens the power of insurance companies over individuals' behavior. There naturally is a power relation between these two parties, but the power relation becomes (much) more asymmetric when insurance providers are able to surveille clients. This raises the question: how much power do we want insurance companies to have over people's behavior? Take the example of video surveillance inside a person's vehicle, aimed at judging a person's driving style. Just like it seems sensible that law enforcement uses its power to make people obey the law, it also seems sensible that insurance companies take efforts to stimulate safe driving. While most people would agree that one should obey the law and drive safely, it still seems wrong to be forced to do so. As already pointed out in the previous section, respecting autonomy requires giving people the freedom to develop and follow their own (moral) judgment.

In the case of workplace surveillance by employers over employees, a similar issue arises. CV also enables employers to automatically surveille employees. Depending on the work environment, workplace surveillance can entail detecting certain behaviors like chatting with co-workers or smiling to customers, or taking record of the time spent on specific tasks or at a desk. Again, it seems sensible for employers to see to it that their employees do their job as they should. However, using the power of CV automated surveillance to get employees to behave as expected, employers violate the ability of employees to follow their own judgment and do well at their job out of their own volition. Moreover, workspace surveillance signals distrust to employees, which can negatively affect their relationship to their employer and the joy or pride employees experience in doing their job.

CV also makes camera security more affordable for private persons and enables new kinds of products, such as systems that monitor baby-rooms and warn the parents when the child wakes up or tries to climb out of their bed; systems that detect when elderly people fall inside their homes and automatically

Table 2 Episodic power

How can CV facilitate the exercise of power?	Does this instance of power affect autonomy?
Stricter enforcement of the law through automated camera surveillance	Challenges ability/freedom to develop and follow one's own (moral) judgment
Disproportional power of insurers over clients due to use of automated camera surveillance	Challenges ability/freedom to develop and follow one's own (moral) judgment
Disproportional power of employers over employees due to use of automated camera surveillance	Challenges ability to follow one's own judgment and to develop a sense of identity in relation to one's work
The exercise of power by members of a household over other members of that household, or by neighbors over other neighbors, using automated camera surveillance	Challenges ability to develop and follow one's own judgment. Violates informational control and therefore challenges ability to own life story
Personalized marketing using CV	Challenges ability to develop and follow own judgment

contact emergency help; or smart doorbells that show you who is in front of your door. These technologies bring surveillance into the private sphere and offer tech companies an insight into people's private lives. Moreover, these video analytics applications change the power dynamics present in social relations. In some cases, this can be empowering. For example, smart home security with fall detection makes elderly people less dependent on caretakers, which can be professionals, but also their family or neighbors. However, smart camera surveillance in the private sphere can also change the power dynamics in social relations in ways that can hurt people's autonomy. For instance: smart camera surveillance gives parents more power over their children, because it enables them to watch their kids even when they are not at home. This gives children less anonymity and less freedom to act as they wish to, both of which children need to develop a sense of identity and autonomy. Another example: smart doorbells not only tell someone who is ringing their doorbell, they often also film the entire lawn or a part of the street and sidewalk. By doing so, smart doorbells provide data of the mail carrier's face or about the neighbor's visitors. Such peer-to-peer surveillance (also known as 'horizontal surveillance') not only violates people's privacy and informational control, it also creates an asymmetrical relation of knowledge and power between neighbors.

In addition to its use for surveillance, CV can also serve business and marketing tools like personalization, that enable companies to influence or manipulate consumer behavior. An example of this is the use of facial recognition to identify and categorize customers, on the basis of which those customers can be targeted with tailored advertisements or deals. Depending on how strong or successful the influence of such targeted or personalized products and services is, we can call this an exercise of power as well. It is problematic when companies have too much power over consumer behavior, because it takes away consumers' ability to make their own, autonomous decisions. See Table 2 for an overview of CV's potential normative implications related to episodic power.

6 Systemic power

In this section, I discuss how CV creates or contributes to systemic power and when this stands in the way of emancipatory progress. The systemic view of power focuses not on individual exercises or instances of power, but on the structural power relations that are reflected in the norms, practices, and institutions that rule a society. Systemic power affects autonomy by determining what opportunities people have, the values people adopt and shape themselves after, and the kinds of choices people make. For instance: the dominance of traditional gender norms in relationships and societal institutions can keep women from pursuing a career. So systemic power is relevant in the context of emancipation, mainly because it plays a crucial role in people's ability to shape their own lives and identities. Just like other aspects of power, systemic power is not necessarily morally problematic. Societal systems are always going to shape people's lives in some way, but emancipation requires that opportunities are available to develop oneself and flourish. Emancipatory progress also requires that people are able to go against certain ways of life, that is, to contest systemic power.

For starters, CV can strengthen the systemic power of institutions such as governments and businesses. CV applications like automated surveillance cameras or bodycams can support governments' systemic power by making law enforcement more efficient and effective. CV can support business intelligence tools or personalized services, which can help businesses to increase profit and improve their market power. Moreover, the companies that develop CV applications increase their market power by selling their products to smaller industries and governments and by acquiring immense amounts of valuable data on people's appearance, expressions, and whereabouts.

When a government, company, or a whole industry has significant systemic power, they are able to determine societal norms and practices. Over the past decades, big tech companies became so powerful that they were collectively able to give rise to a new mode of capitalism: surveillance

capitalism [26]. The surveillance capitalist business model consists of the datafication and commodification of people's behavior as they interact with digital technology. So far, surveillance capitalists mainly datafied and commodified people's online behaviors (such as search queries and online shopping behavior). However, CV can contribute to the rise of surveillance capitalist practices outside of the online sphere, by datafying people's appearance: from the way they walk to the way they dress, and from facial recognition to emotion recognition. The market power of big tech companies makes it difficult to compete with their businesses and makes it challenging for individuals to escape or contest surveillance capitalism. Furthermore, surveillance capitalists have used their systemic power to reify their practices, by leading people to believe that there is no alternative. Despite concerns about privacy and data-ownership, surveillance capitalist practices have been more or less accepted by societies. One could say (as Zuboff does) that this acceptance stems from the fact that people are led to believe that there is no alternative to surveillance capitalism. People were made to believe that, if they want to enjoy (free) digital services, they need to accept the datafication and commodification of their behaviors and characteristics. This reification also prevents the masses from questioning, criticizing, or countering the status quo. This systemic power of big tech companies and surveillance capitalism can stand in the way of people's self-legislation.

Another reified systemic power relation between user and developer is the fact that tech companies usually exercise full authority over the development of the technologies that come to dominate nearly everyone's lives and societies. It does not have to be this way. Alternatively, individuals or policy makers are given a much stronger say in which technologies are developed and the details of their design [16]. The systemic power of big tech companies to determine which (CV) technologies come to dominate people's lives and societal practices, stands in the way of citizens' collective self-legislation.

Furthermore, CV applications sometimes reflect and even exacerbate pre-existing systemic societal injustices. A vivid example of this is the problem of algorithmic bias. Particularly infamous examples of algorithmic bias are biased facial recognition systems, that have often shown to be more likely to recognize White people and males than darker skinned persons and females (e.g. [3]). These biases have led to false arrests by law enforcements agencies in the US that used facial recognition to detect criminal offenders, but also to countless demeaning cases where Black people were categorized as 'gorillas' or 'apes' or Asian people as constantly 'blinking'. The categories used by facial recognition systems also often confirm problematic social stereotypes. For example, darker skinned people are more likely to be given a label associated with crime or violence. Also, many gender categorization systems are binary, which is obviously a problem when the subject in question identifies as non-binary. By maintaining or exacerbating systemic injustices, CV systems can harm people's self-development. Self-development is compromised either because algorithmic biases keep people from exercising certain freedoms or getting equal opportunities (e.g. when biased systems are involved in hiring practices), or because they harm a person's sense of self-worth (e.g. when falsely recognized as criminal by law enforcement or when given demeaning labels by personal devices).

CV applications also reinforce existing societal power structures to the extent that they normalize certain behaviors, views, or identities. Based on visual data, among others, AI systems can stimulate particular behaviors and recommend specific courses of action. For example, semi-automated vehicles can recommend certain actions based on what is happening on the road, which in turn can normalize rather safe or rather dangerous driving styles. Similarly, augmented reality applications could normalize specific behaviors by suggesting courses of action in response to the real-world environment (e.g. "there's your neighbor, say hi!"). Furthermore, CV systems categorize people in accordance with demographic information such as gender, age, race or religion, but also according to other characteristics that one's appearance might reveal, like 'shy', 'poor', 'fashionable', and so on. These labels could reflect and confirm societal stereotypes and norms. Take the aforementioned example of binary gender categorization systems, these reflect and reinforce the norm that a person is either male or female (and, arguably, that it is important to categorize people by gender). Normalization by CV applications can hinder people from developing and following their own judgment, as well as hindering the development of an own sense of identity.

Another issue of systemic power is CV's impact on the labor market. First of all, CV, like other technologies, could automate tasks that make certain jobs and professional skills superfluous. Think about the human operators that used to be needed to monitor CCTV footage. While history shows that jobs that are automated are always replaced by new kinds of jobs, automation might still be a serious threat to individuals whose skills become irrelevant on the labor market and who struggle to find other ways to make a living. Secondly, CV has developed immensely in the last decade or so due to the availability of ImageNet—a large database with labeled images that can be used to train CV models on. ImageNet, in turn, is made possible by the use of crowd sourced workers who label images in return for a small compensation. In both cases it is the scale of the data needed to develop CV and the scale with which CV is and will be implemented, that makes CV able to impact jobs or labor at a systemic, market level.

Finally, the environmental cost of data storage and processing is an important matter of systemic power as well. As Crawford writes: "The massive ecosystem of AI relies

Table 3 Systemic power

How does CV tie into systemic relations or structures of power in society?	How do these systems of power impact people's autonomy?
CV could extend the reach of surveillance capitalist practices and thus increase the systemic power of surveillance capitalists	Limiting people's ability to escape surveillance capitalist practices or to counter or criticize their role in everyday life and society
Tech companies determine which CV applications come to dominate people's lives and societies	Limiting collective self-legislation of citizens
Biased CV algorithms	Limiting self-development because of a lack of equal opportunities or harms to self-worth
CV normalizes certain behaviors and identities	Limiting formation of own judgment and own identity
CV is part of a system of power that has significant environmental costs	Violates human self-legislation to the extent that activist individuals or groups do not stand a chance when trying to counter the environmental impact of AI and big data
CV can cause lasting changes in specific job markets	Limits the self-development and self-legislation of those forced to re-educate themselves, change industries, or work for an exploitative wage

on many kinds of extraction: from harvesting the data made from our daily activities and expressions, to depleting natural resources, and to exploiting labor around the globe so that this vast planetary network can be built and maintained." [6, p. 32]. CV, like other forms of data analytics, relies on big data. Therefore, CV contributes to the environmental damage done by the processing and storage of big data. This is an issue of systemic power: the growing societal dependence on AI maintains a system of power between humans and the environment, that cannot be fought by those trying to protect the environment. The central harm here is obviously environmental harm, but the systemic power of the AI industry simultaneously violates human autonomy, because activist individuals or groups do not stand a chance when trying to counter the environmental impact of AI and big data. See Table 3 for an overview of the issues discussed in this section.

7 Constitutive power

Constitutive power is a view of power that is often associated with the work of Foucault. Constitutive power concentrates not on having or exercising power, but on the effects of power on those subjected to it. Moreover, it looks not at the oppressive side of power (on the fact that A's power keeps B from doing x), but on the creative side of power (namely the fact that A's power makes B do y instead of x). Because of this focus, the constitutive view of power complements the aforementioned aspects of power. Looking at the constitutive aspect of power is ethically relevant, because the ways in which we are moved to act, think, or shape our identity can be normatively laden. Being aware of this normativity enables one to criticize it.

A first way in which computer vision constitutes our behavior, thoughts or identity is through surveillance. I already touched upon this issue above. Surveillance cameras usually represent a certain authority—the state, the security staff of a shopping mall, an employer, one's parents, and so on. Upon seeing a surveillance camera, a person might alter their behavior out of respect for the authority or because of the threat of repercussions (e.g., getting a fine or getting fired). But surveillance cameras not only cause people to alter their behavior in particular instances, they can also lead to the long-term internalization of official rules or social norms of conduct. Hence, computer vision enforces certain behavioral norms (as also pointed out in the previous section). Although the enforcement of behavioral norms can help to establish a safer environment, it can also support undesirable or unreasonable norms about being a good citizen, employee, child, etc., that are too restrictive with regards to people's ability to form and follow their own judgment and constitute their own identity.

Second, computer vision is not yet a much used technique in the context of personalization. However, it has the potential to provide effective input for personalization, by analyzing a person's appearance (from clothing style to facial expression) and behaviors (from the way a person walks to the things they focus their eyes on). Computer vision namely makes it possible to make numerous inferences about a person's identity, behavior, and other characteristics, on the basis of which that person can be offered personalized offers, messages, goods, or services. Potentially, it would be able to reveal information about a person that they are not even aware of themselves. Like surveillance, personalization is aimed at nudging or manipulating people's behavior. For instance, personalization could move consumers to buy things they otherwise would not buy or encourage voters to alter their political views. Personalization is inherently normative, because it always promotes a certain product, article, political party, course of action, and so on, as being the best

Table 4 Constitutive power

How can CV applications constitute people's behavior, thoughts, and identity?	How do these constitutive effects impact people's autonomy?
CV surveillance causing the internalization of specific norms	Limiting formation of own judgment and own identity
CV can support personalization	Limiting ability to follow and form own judgment (self-legislation)
Labels assigned by CV can constitute a person's sense of identity	Potentially harmful to self-development
Discomfort of being watched or analyzed can cause behavioral changes	Potentially harmful to ability to act freely

choice for a person. In doing so, it not only shapes specific choices people make (e.g. which shoes to buy or which song to listen to), it can shape their preferences and views in the long term. Recommendation systems can shape our political views, our preferences, and even our beliefs about the world and about truth, which is harmful to people's self-legislation.

Thirdly, some computer vision applications, in particular facial recognition and analysis systems, are aimed at identifying and categorizing people on the basis of their appearance. Such applications can be used in numerous ways: for example to detect suspicious people in an airport, to identify the type of clientele in a store, to personalize products online, or to create fun face filters on social media. The labels a person is given by such tools shape how they are perceived by the world around them, but also shape how they grow to understand and develop themselves. Computer vision can therefore also have constitutive effects on a person's identity and self-development.

Finally, the presence of a camera, whether it serves surveillance or other purposes, creates a sense of constantly being watched. Computer vision could exacerbate this feeling, as automated analysis is more rigorous than a human eye. To some, the sense of being watched is so uncomfortable that it leads them to alter their behavior. This issue is sometimes referred to as 'chilling effects'. However, while 'chilling effects' implies that we refrain from acting at all (e.g. refrain from exercising rights, such as participating in a protest), the discomfort of being watched can also mean we simply behave differently. So computer vision applications can make us change our behavior in ways that were initially not intended by those who developed or implemented the technology. This means that, even when a camera was not meant to make us change our behavior, it still has constitutive power over our behavior. The bigger the discomfort people experience when 'watched' by smart cameras, the bigger the impact on people's ability to act freely. See the summary of potential issues related to constitutive power in Table 4.

8 Conclusion

The aim of this article was to provide an overview of the potential normative implications of CV. Given the wide variety of possible CV applications, the focus of the analysis was predominantly on those CV applications that involve the analysis of persons. To create this overview I applied a critical framework, centered around a pluralistic understanding of power. I identified power dynamics related to CV applications and argued how those power dynamics could impact people's autonomy. Considering how CV might harm autonomy is important to ensure that the technological progress, at least in the field of CV, goes hand in hand with social, emancipatory progress.

A limitation of this approach is that it does not necessarily offer guidance in determining how severe an application's impact on autonomy will really be. Rather, the framework points out *potential* impacts on autonomy, that should each be studied in more detail in practice, to determine the severity of the issue and the possible routes of countering the impact on autonomy and, thus, improving the technology. However, despite this limitation, I conclude that the overview forms a valuable contribution to the literature on AI ethics and on computer vision.

Acknowledgements The author is an early stage researcher funded by MSCA ITN PROTECT, H2020 grant agreement no. 813497.

Declarations

Conflict of interest The author declares that there are no conflicts of interest.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Blank, E.L., Weltz, E., Arias, C.: Computer Vision machine learning and future-oriented ethics. *Honors Projects.* 107 (2019). https://digitalcommons.spu.edu/honorsprojects/107
2. Brey, P.: Ethical aspects of facial recognition systems in public places. J. Inf. Commun. Ethics Soc. **2**(2), 97–109 (2004). https://doi.org/10.1108/14779960480000246
3. Buolamwini, J., Gebru, T.: Gender shades: intersectional accuracy disparities in commercial gender classification. Conference on Fairness, Accountability, and Transparency, pp. 77–91 (2018)
4. Coeckelbergh, M.: AI ethics. MIT Press (2020)
5. Coupland, S., Wakunuma, K., Stahl, B.: Identifying ethical issues during the development of a computer vision based AmI system: A case study. Roots for the Future of Ambient Intelligence - Adjunct Proceedings, 3rd European Conference on Ambient Intelligence, AmI 2009, May 2014, pp. 290–297 (2009)
6. Crawford, K.: Atlas of AI: power, politics, and the planetary costs of artificial intelligence. Yale University Press (2021)
7. Crawford, K., Paglen, T.: Excavating AI: The politics of Images in Machine Learning Training Sets. excavating.ai. (2019). Accessed 31 Oct 2022.
8. Doyle, T., Veranas, J.: Public anonimity and the connected world. Ethics Inf Technol. **16**, 207–218 (2014). https://doi.org/10.1007/s10676-014-9346-5
9. Dubber, M.D., Pasquale, F., Das, S. (eds.): The Oxford Handbook of Ethics of AI. Oxford University Press (2020). https://doi.org/10.1093/oxfordhb/9780190067397.001.0001
10. Dufresne-Camaro, C., Chevalier, F., Ishtiaque Ahmed, S.: Computer Vision Applications and their Ethical Risks in the Global South. Graphic Interface Conference. (2020). https://openreview.net/pdf?id=QLFSDNIvI
11. Gordon, J. S., Nyholm, S.: Ethics of Artificial Intelligence. Internet Encyclopedia of Philosophy. (2021). https://iep.utm.edu/ethic-ai/
12. Hale, B.: Identity crisis: face recognition technology and freedom of the will. Ethics Place Environ. **8**(2), 141–158 (2005)
13. Haugaard, M.: Power: a "family resemblance concept." Eur. J. Cult. Stud. **13**(4), 419–438 (2010)
14. Huffer, D., Wood, C., Graham, S.: What the Machine Saw: some questions on the ethics of computer vision and machine learning to investigate human remains trafficking. Internet Archaeol. (2019). https://doi.org/10.11141/ia.52.5
15. Lauronen, M.: Ethical issues in topical computer vision applications. Information Systems, Master's Thesis. Jyväskylä: University of Jyväskylä. (2017). https://jyx.jyu.fi/bitstream/handle/123456789/55806/URN%3aNBN%3afi%3ajyu-201711084167.pdf?sequence=1&isAllowed=y
16. Maas, J.: Machine learning and power relations. AI & Soc. (2022). https://doi.org/10.1007/s00146-022-01400-7
17. Mitchell, M.: Artificial Intelligence: a guide for thinking humans. Pelican Books, London (2019)
18. Moraes, T.G., Almeida, E.C., de Pereira, J.R.L.: Smile, you are being identified! Risks and measures for the use of facial recognition in (semi-)public spaces. AI Ethics (2020). https://doi.org/10.1007/s43681-020-00014-3
19. Morriss, P.: Power: a philosophical analysis. Manchester University Press, Manchester, New York (2002)
20. Müller, V.: Ethics of Artificial Intelligence and Robotics. In: Zalta, E. N. (ed.). The Stanford Encyclopedia of Philosophy. 1st edn (2020). https://plato.stanford.edu/entries/ethics-ai/
21. Sattarov, F.: Power and technology: a philosophical and ethical analysis. Rowman and Littlefield International, London, New York (2019)
22. Selinger, E., Leong, B.: The ethics of facial recognition technology. In: Véliz C (ed) The Oxford Handbook of Digital Ethics (forthcoming). https://doi.org/10.1093/oxfordhb/9780198857815.013.32
23. Skirpan, M., Yeh, T.: Designing a moral compass for the future of computer vision using speculative analysis. In: IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), pp. 1368–1377. (2017). https://doi.org/10.1109/CVPRW.2017.179
24. Stanley, J.: The Dawn of Robot Surveillance. AI, Video Analytics, and Privacy. American Civil Liberties Union. (2019). https://www.aclu.org/sites/default/files/field_document/061119-robot_surveillance.pdf
25. Waelen, R.: Why AI ethics is a critical theory. Philos. Tech. (2022). https://doi.org/10.1007/s13347-022-00507-5
26. Zuboff, S.: The age of surveillance capitalism: the fight for a human future at the new frontier of power. Profile Books, London (2019)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Philosophy & Technology (2022) 35: 9
https://doi.org/10.1007/s13347-022-00507-5

RESEARCH ARTICLE

Why AI Ethics Is a Critical Theory

Rosalie Waelen[1]

Received: 20 October 2021 / Accepted: 1 February 2022 / Published online: 11 February 2022
© The Author(s) 2022

Abstract
The ethics of artificial intelligence (AI) is an upcoming field of research that deals with the ethical assessment of emerging AI applications and addresses the new kinds of moral questions that the advent of AI raises. The argument presented in this article is that, even though there exist different approaches and subfields within the ethics of AI, the field resembles a critical theory. Just like a critical theory, the ethics of AI aims to diagnose as well as change society and is fundamentally concerned with human emancipation and empowerment. This is shown through a power analysis that defines the most commonly addressed ethical principles and topics within the field of AI ethics as either to do with relational power or with dispositional power. Moreover, it is concluded that recognizing AI ethics as a critical theory and borrowing insights from the tradition of critical theory can help the field forward.

Keywords Artificial intelligence · AI ethics · Critical theory · Power

1 Introduction

The ethics of artificial intelligence (AI) is an emerging field within applied ethics and the philosophy of technology. It has gained attention and urgency due to the rapid development of AI technology during the past decade. Following Müller (2020), I understand the purpose of the field as twofold: AI ethics deals with the ethical issues that arise from AI systems as objects and with the moral questions raised by AI systems as subjects.[1] Various approaches for the ethical analysis of AI systems as objects have been proposed, but the most dominant one appears to be the *principled approach*. Over the past years, numerous initiatives have developed comparable sets of ethical principles and guidelines to ensure a desirable development and use of AI (Jobin et al., 2019; Ryan & Stahl, 2020). Common principles

[1] For an overview of the field see Dubber et al., 2020; Gordon & Nyholm 2021; or Müller 2020.

✉ Rosalie Waelen
r.a.waelen@utwente.nl

[1] Section of Philosophy, University of Twente, Enschede, Netherlands

are for example transparency, justice and fairness, non-maleficence, responsibility, and privacy (Jobin et al., 2019). These principles function as a kind of soft law, they are not binding. The moral questions that AI systems raise as subjects are, among others, according to what norms AI should be programmed to act (also referred to as "machine ethics"), what the moral status of artificial moral agents would be, and whether AI systems can be held responsible or accountable for their actions and decisions.

The purpose of this article is to argue that AI ethics has the characteristics of a critical theory, by showing that the core concern of AI ethics is protecting and promoting human emancipation and empowerment. Furthermore, I propose that understanding the field as a critical theory can help to overcome some of the shortcomings of the currently popular principled approach to the ethical analysis of AI systems. In other words, I argue that we not only could, but should analyze the ethical implications of AI systems through the lens of critical theory. The focus of the article therefore lies on the ethical analysis of AI systems and the principled approach, but I also discuss what a critical understanding of AI ethics means for the other topics and questions that make up the field.

The structure of the article is as follows. In Sect. 2, I start by briefly defining what it takes to be a critical theory. I conclude that a critical theory studies the power structures and relations in a society, with the goal of protecting and promoting human emancipation, and seeks not only to diagnose, but also to change society. In Sect. 3, I take a more detailed look at the concept of power and explain how a pluralist understanding of the concept allows us to analyze power structures and relations, on the one hand, and (dis)empowerment, on the other hand. In Sect. 4, I argue that the vast majority of the established AI ethics principles and topics in the field are fundamentally aimed at realizing human emancipation and empowerment, by defining these issues in terms of power. Next, in Sect. 5, I propose that AI ethics should be seen as a critical theory, given that the discipline is fundamentally concerned with emancipation and empowerment, and meant not only to analyze the impact of emerging technologies on individuals and society, but also to change it. Moreover, I suggest that recognizing AI ethics as a critical theory can help the field forward—among other reasons because it promotes an interdisciplinary understanding of ethical issues, offers a target for change, helps to identify social and political implications of a technology, and helps to understand the social and political roots of ethical issues. I end the chapter with a brief conclusion in Sect. 6.

2 What Constitutes a Critical Theory?

Marx' famous thesis that philosophers should not only interpret the world but also change it, inspired a group of philosophers now known as the Frankfurt School.[2] The work of the Frankfurt School philosophers was given the name "critical theory."

[2] This is the 11th thesis on Feuerbach which Karl Marx wrote in 1845. It was later published as an appendix to Friedrich Engels' *Ludwig Feuerbach and the End of Classical German Philosophy* in 1886.

Critical theory has two main, characteristic facets. First of all, critical theory has a practical goal; it is meant to diagnose as well as change society. Its unique approach to this is that of immanent transcendence, which implies that critical theorists argue how the world should be (transcendence) based on how it currently is (immanence), rather than always working towards a single predefined image of an ideal state of affairs (Delanty & Harris, 2021; Thompson, 2006). Horkheimer, who founded the Frankfurt School together with Adorno in the early 1930s, suggested that critical theory should be an interdisciplinary endeavor. History, economics, politics, psychology, and other social sciences can help to understand in what ways people's freedom is limited, how the power relations causing this domination came about, and how to counter or resist them. A second important facet is critical theory's emancipatory ambition. Horkheimer said critical theory is "an essential element in the historical effort to create a world which satisfies the needs and power of men" and defined its goal as "man's emancipation from slavery" (Horkheimer, 1972, 246). So critical theorists always seek to identify and overcome forms of domination or restraints that hinder human emancipation or empowerment. Emancipation can be defined as "overcoming social domination" (Forst, 2019, 17) and gives people an equal opportunity for self-development (Allen & Mendieta, 2019). Empowerment implies "[i]ncreasing the scope of agency for individuals and collectives" (Forst, 2019, 21).

There are distinct generations within critical theory. The first generation of critical theorists (among which were Theodor Adorno, Walter Benjamin, Max Horkheimer, and Herbert Marcuse) was preoccupied with criticizing modern capitalism and discussed typically Marxist subjects like alienation, exploitation, and reification. Later on, the focus of their critique became the enlightenment and the loss of individuality due to mass culture (Horkheimer & Adorno, 2002). Jürgen Habermas, a second-generation critical theorist, continued the tradition by studying the state of democracy and discussing power in relation to communication, which lead him to develop his discourse ethics (Habermas, 1984, 1987). Axel Honneth, a student of Habermas, in turn focused his attention on the topic of recognition, which goes back to Hegel (Honneth, 1996). One of the contemporary, fourth-generation members of the school is Rainer Forst, who has continued the tradition by developing a critical theory of justice and redefining the notions of progress and power, among others.

Only the first generation of critical theorists explicitly concerned themselves with technology, mostly focusing on its relation to capitalism (Delanty & Harris, 2021). But, the types of technology that these early Frankfurt School members dealt with were nothing like AI and other digital technologies that exist today. Therefore, it is not easy and perhaps not very valuable either, to try to apply their theories of technology to today's situation. However, that does not have to mean that the tradition of critical theory is not relevant to the philosophy and ethics of technology. Several contemporary thinkers have argued for the relevance of critical theory to understanding the societal role and impact of technology today. Most notably, Feenberg engaged with this tradition to develop his own critical theory of technology (a.o. Feenberg, 1991). Another example is Fuchs, who built on the work of Lukács, Adorno, Marcuse, Honneth, and Habermas to develop a critical theory of communication in the age of the internet (Fuchs, 2016). And in a recent article, Delanty and Harris argue that the general themes that are present in critical theory still offer a

valuable framework for analyzing technology today (Delanty & Harris, 2021). So, the central idea of this paper, namely that the tradition of critical theory can support the analysis of modern technology, is not necessarily new. What is new, as will become clear in what follows, is my proposal to understand the emerging field of AI ethics as a critical theory and to conduct ethical analyses of AI systems through the lens of critical theory.

Finally, the understanding of critical theory as being the work of the Frankfurt School is a narrow understanding of the term. When understood in a broader sense, the term "critical theory" can refer to any theory or diagnosis of power structures and relations that ought to serve emancipatory ends. In this broad sense, the critical theory would also include other schools of thought, such as feminism or postcolonialism (Bohman, 2021). We could then say that specific critical approaches or theories focus on a particular oppressed societal group or a particular way in which people's emancipation is hindered. Different critical theories deal with the struggle of a specific day and age (Bohman, 2021). AI ethics is a critical theory, as I further argue below, which deals with the ways in which AI—a radically new technology—(dis)empowers individuals and facilitates or exacerbates existing power structures in society.

3 Defining the Concept of Power

Given critical theory's focus on emancipation and empowerment, and all the factors that enable or disable this, we can conclude that power is an important, central topic within the field of critical theory. However, power is also a contested concept—according to Steven Lukes it even is essentially contested (Lukes, 1974, 2005). Therefore, it is difficult to understand power's exact role in critical theory. When trying to define "power," scholars disagree over a number of issues. For example, it is disputed whether power can be ascribed to structures or solely to agents; whether an exercise of power is necessarily intentional or whether someone or something can exercise power without intending to do so, and whether the exercise of power has to involve a conflict of interests or not (Brey, 2008). Furthermore, there is a divide between those who conceptualize power in dispositional terms, as "power-to," and those who discuss power in relational terms, as "power-over."

We could say that there are, broadly, four different views of power: the dispositional, episodic, systemic, and constitutive view (Allen, 2016; Haugaard, 2020; Sattarov, 2019). The first resembles "power-to," the latter three are relational views, and therefore, they fall under the category of "power-over." Those who defend the dispositional view of power argue that power is a capacity or ability—namely the capacity to bring about significant outcomes. Acquiring that capacity is also referred to as "empowerment," losing it as "disempowerment." One defender of the dispositional view is Morriss, who argued that two mistakes are commonly made in discussions of power: the vehicle fallacy and the exercise fallacy (Morriss, 2002). The vehicle fallacy is committed when the resources that give rise to power (e.g., AI technologies) are claimed to be power. The exercise fallacy occurs when one equates power with its exercise. Morriss argues that having power entails more than merely

exercising it, it is a disposition that "can remain forever unmanifested" (2002, 17). This is then a direct critique towards those who defend a relational view of power, in particular the episodic view.

The episodic view of power entails that power occurs when one party exercises power over another, for example, by means of force, coercion, manipulation, or through authority. Known for defending, this view of power is Weber, Dahl, and Lukes. Dahl famously formulated the intuitive notion of power as "A having power over B to the extent that A can get B to do something that B would not otherwise do" (1957, 202). Lukes initially followed Dahl, by defining power as "A exercises power over B when A affects B in a manner contrary to B's interests" (1974, 30), but later in his career he accepted Morriss' critique and acknowledged that power is "a capacity not the exercise of that capacity" and that "you can be powerful by satisfying and advancing others' interests" (Lukes, 2005, 12). But even though dispositional power appears to be more fundamental than episodic power, the episodic view of power is relevant because it highlights a specific aspect of power, namely the direct exercise thereof.

Other relational views of power are the systemic and constitutive view. While dispositional and episodic power focus on a single agent and specific instances of power, systemic and constitutive power are more structure-centric (Allen, 2016; Sattarov, 2019). Systemic power, to start with, refers to the ways in which societal institutions, social norms, values, laws, and group identities can have power over individuals. The systemic view complements the episodic and dispositional ones, because it enables us to look at the bigger picture and see what causes some to have dispositional power and exercise it, while others cannot or are constantly subjected to the power of others. Systemic power "highlights the ways in which broad historical, political, economic, cultural, and social forces enable some individuals to exercise power over others, or inculcate certain abilities and dispositions in some actors but not in others." (Allen, 2016).

Constitutive power, finally, refers to the views or discussions of power that focus not on the oppressive character of power, but on the ways in which those subjected to power are also shaped by it. Systems of power not only determine one's sphere of action or possibilities, as the systemic view of power highlights, they also constitute a person's behavior, intentions, beliefs, and more. Foucault is probably most famous for developing this view of power in his work on discipline and biopolitics.

These four views of power are not necessarily incompatible, competing views. According to Mark Haugaard (2010, 2020), Lukes is wrong in saying that power is essentially contested. Haugaard suggests that "power debates will advance more fruitfully if we treat power as a *family resemblance* concept, whereby their meaning varies depending upon *language game*" (Haugaard, 2010, p. 424).[3] A family-wide concept like power is so broad and vague that it explains little in itself; therefore, it is better understood through a cluster of concepts that refer to different aspects of the wider notion. Hence, we should reject the premise that there is a single best definition of power to be found and opt for a pluralist approach to power. The criterion for

[3] Both the idea of family resemblance and language games are borrowed from Wittgenstein.

including a certain view of power in the pluralist approach should be "usefulness," says Haugaard (2010, 427). A definition or notion of power is useful, when it highlights a unique aspect of power.

Critical theorists are concerned with all four elements of power. They first and foremost study relational power, particularly systemic issues in society, in order to emancipate certain societal groups. But, critical theorists are also interested in increasing the scope of human agency, that is, empowering individuals and groups. This latter concern ties in with dispositional and constitutive power. Hence, all four notions of power are valuable in order to understand AI ethics as a critical theory and to conduct ethical analyses of AI systems through the lens of critical theory.

4 Analyzing AI Ethics Principles and Topics in Terms of Power

As explained in the introduction, the field of AI ethics is concerned with identifying and addressing the ethical implications of AI as well as the moral questions raised by this new technology. Müller (2020) discusses privacy, manipulation, opacity, bias, the future of work, and autonomy as main ethical issues that arise from AI systems as objects, and mentions machine ethics, artificial moral agency, and singularity as topics to do with AI systems as subjects. Gordon and Nyholm (2021) offer a similar list. As main debates in the ethics of AI they name machine ethics, autonomous systems, machine bias, opacity, machine consciousness, moral status, and singularity. To some extent, these lists overlap with sets of AI ethics principles or guidelines. It has namely become a trend, among academics as well as businesses and policymakers, to develop sets of ethical principles that should guide the development and use of AI in a desirable direction. This trend represents the aforementioned principled approach to AI ethics.[4] A comparative study of 84 sets of AI ethics principles showed that there is a lot of convergence between the principles that different parties have proposed (Jobin et al., 2019). More precisely, the study identified eleven clusters of values and principles that were brought forward in several documents: transparency (mentioned in 73 of the studied documents), justice and fairness (68), non-maleficence (60), responsibility (60), privacy (47), beneficence (41), freedom and autonomy (34), trust (28), sustainability (14), dignity (13), and solidarity (6). These principles touch upon the debates that are central in AI ethics according to Müller (2020) and Gordon and Nyholm (2021). The issue of opacity, for example, relates to the principle of transparency, and the issue of bias is addressed by the principle of justice.

In what follows I define the most-mentioned AI ethics principles (Sects. 4.1, 4.2, 4.3, 4.4, 4.5, 4.6, and 4.7) in terms of power. By doing so, I show that the fundamental concerns that underly these principles are emancipation, empowerment, or both. Furthermore, I also discuss how other topics in AI ethics, such as machine ethics or

[4] The principled approach to AI ethics resembles the dominant approach in the more established field of bioethics, where four ethical principles are widely recognized as the basis for policymaking and clinical decision-making in the medical field.

singularity, relate to the concept of power (Sect. 4.8). In the last section (Sect. 4.9), I briefly look at non-Western approaches to AI ethics and argue that the concern for emancipation and empowerment is present (although perhaps less dominant) in these approaches as well.

Finally, when I talk about AI as having or exercising power over humans, I do not mean to ascribe the technology as the kind of agency or intentionally required to purposefully seek power. Rather, I merely mean that technology can be used as an instrument or delegate by human beings to exercise power (Brey, 2008) or that AI can unintentionally create new power relations, exacerbate existing ones, or affect an individual's autonomy by empowering or disempowering them.[5]

4.1 Transparency

The most proposed principle for ethical AI is the principle of transparency. Transparency would be required, for example, with regard to data collection and processing (what is done with my personal information?), automated decision-making (how are the decisions that affect me made?), or personalized recommender systems (on the basis of what data am I shown a certain product or news article?). Transparency implies that the answers to these questions are both accessible and comprehensible. The idea behind this principle is that transparent, explainable, or interpretable AI would minimize harm by AI systems, improve human-AI interaction, advance trust in the technology, and potentially support democratic values (Jobin et al., 2019). A different way of explaining what makes transparency in AI so valuable is to say that it grants an individual epistemic agency, i.e., the ability to know what happens to their data and how AI affects them, as well as the ability to better control one's own information and the ways in which they are affected by AI. This knowledge and control are dispositional powers—transparency implies having the power to know or understand what happens to one's data and on what bases decisions are made. Therefore, we can say that what is really at stake when transparency is called for, is individual empowerment.

[5] It should also be noted that there exist a number of examples of discussions of power within AI ethics or related fields. For instance, Cobbe (2020) argues that algorithmic censorship augments the societal power of social platforms; Danaher (2020) discusses the effects of "algocracy" on freedom; de Laat (2019) defends that predictive algorithms have disciplinary power; Mohamed et al. (2020) suggest the use of decolonial theory to understand how the values embedded in AI are shaped by existing power relations; Noble (2018) discusses the "oppressive algorithms" of search engines; and Zuboff (2019) critiques the dominance of "surveillance capitalism." Also, Sattarov (2019) explores the relation between technology (e.g., algorithms), power and ethics. So there has certainly been recognition for the relevance of discussing power in relation to ethical issues in AI, but these issues are not covered or addressed by the dominant AI ethics guidelines. Moreover, each of the mentioned discussions of power focuses on a single ethical issue, a single technology, or a specific conception of power. The critical approach to AI ethics that will be put forward in this paper, which is based on a pluralist understanding of power, brings these different issues together.

4.2 Justice, Fairness, and Solidarity

Justice is mentioned in AI ethics guidelines in relation to fairness, on the one hand, and bias and discrimination, on the other hand (Jobin et al., 2019). Fairness concerns have to do with equal access to AI and the equal share of the burdens and benefits of the technology. There is, for example, the concern about a digital divide between the countries that can afford to develop and use AI and those parts of the world that do not have access to the latest technology. The principle of non-discrimination has become pressing as many emerging technologies have been found to contain biases. Particularly algorithmic bias has received much attention in the field of AI ethics. Algorithms can contain biases, among other reasons, when they are built on non-inclusive training data. Biased algorithms or in other ways biased AI systems can lead to discriminatory outcomes (e.g., continuously misidentify certain demographics as a threat or potential criminal) and therefore violate the principle of just and fair AI. Mentioned less often in existing AI guidelines is the principle of solidarity. Solidarity has to do with the (fair) distribution of AI's benefits and harms. Solidarity in AI would imply that the benefits of AI should be redistributed from those who are disproportionally benefitted by this new technology to those who turn out to be most vulnerable to it (e.g., those who are unemployed due to automation).

These justice-related concerns can be described in terms of relational power. Forst writes that "the question of power, qua social and political power that shapes collective processes, is central to justice" (Forst, 2015, 8). When we speak of justice, we are referring to what we consider to be acceptable power relations or systems in society. The concept of justice can refer to the direct exercise of power by one actor over another (i.e., episodic power), but most often justice has to do with the systems of power that shape a society and, hence, determine the possibilities of certain individuals or groups in that society (systemic power). In other words, the principle of justice protects those subjected or vulnerable to systems of power or the exercise of power by others. In the context of AI, the principle of justice ought to see to it that the power relations that are created or reinforced by AI systems give people the space to develop themselves, and do so in an equal manner. By doing so, the principle of justice serves the goal of emancipation.

4.3 Non-maleficence and Beneficence

Although they are perhaps less strongly related to the concept of power than other principles, non-maleficence and beneficence too can be understood as principles meant to protect those who are vulnerable to the power of AI. As AI develops to be omnipresent in society, it becomes inescapable for people to use the technology, to be subject of data analysis, and to be affected by automated decision-making systems. Their lives are therefore increasingly controlled by this technology. Moreover, the fact that AI is inescapable to the modern citizen is in itself an

instance of systemic power. Protecting that AI does not harm and potentially even benefits its users and subjects is a way of ensuring a desirable power relation and, hence, that AI does not stand in the way of people's emancipation.

4.4 Responsibility and Accountability

Responsibility is frequently mentioned as a guiding principle for AI, because of the concern that automated decision-making will create responsibility gaps (Gordon & Nyholm, 2021). AI systems can take decisions that directly impact human beings, but cannot be held responsible or accountable for the consequences of their actions in the same ways humans can. This raises the question: who should (and can) be held responsible and accountable for the harm caused by AI systems? This question needs to be answered in order to assure that the power relations that AI creates, exacerbates or facilitates are not abused (Sattarov, 2019). Like justice, the principles of responsibility and accountability in AI ethics guidelines can be understood as a way of protecting those subjected to AI's power. Therefore, responsibility too supports emancipation—it helps to overcome unjustifiable social domination.

4.5 Privacy

Privacy can be discussed as a value, moral right or legal right. Jobin et al. (2019) point out that AI ethics guidelines discuss privacy both as a value to uphold and a right that should be protected. Moreover, privacy is often discussed in relation to data protection, which is in line with the common definitions of privacy as "informational control" or "restricted access" (DeCew, 2018). Under both definitions, privacy is understood as a dispositional power, more precisely, as the capacity to control what happens to one's information and to determine who has access to one's information or other aspects of the self. AI, then, is perceived as a potential threat to this capacity because it entails the collection and analysis of large quantities and new types of personal data. Hence, AI could disempower individuals with respect to their privacy. Or put differently, privacy should be promoted because it empowers data subjects.

4.6 Freedom and Autonomy

Freedom and autonomy are related concepts and often mentioned together in AI ethics guidelines. Freedom can be defined in positive and negative terms; it can be understood as the lack of outside interference in one's actions, or the possibility thereof, but it is also discussed as being free to act. The concept of autonomy relates to the positive definition of freedom, it means "self-rule" or "self-determination." If empowerment entails increasing the scope of individual or collective agency, then autonomy and positive liberty clearly serve the goal of empowerment. Having the ability or power to act freely and rule oneself support one's agency. This ability can be promoted, in the context of AI or other types of technology, by for example transparency or informed consent. Negative freedom can be defined in terms of episodic

power and systemic power—it is the absence of power exercises by others or of systemic power relations. One is free, in this sense, when he or she is not subjected to the power of others or a system of power. Such freedom implies, for example, not being subject to technological experiments, manipulation, or surveillance (Jobin et al., 2019). In this sense, the principle of freedom serves the goal of emancipation. We want the values of freedom and autonomy to guide the development and use of AI, because we want to ensure that this new technology is emancipatory and empowering.

4.7 Trust

The call for trust or trustworthy AI can refer to the ways in which AI research and technology is done, to the organizations and persons that develop AI, to the underlying design principles, or to users' relation to a technology (Jobin et al., 2019). Such trust can be fostered by transparency or by ensuring that AI meets the expectations of the public. People's need to trust how AI is developed and functions, can be explained not in terms of dispositional power or empowerment, as is the case for transparency, but as a protection against the exercise of power by others. Trust is a desirable feature of the relation between technology and those using it or subjected to it. When trusting an AI system, one expects that the power that technology can exercise over the individual will not be misused. A trustworthy power relation is one where A holds power over B, without B needing to worry that A will take advantage of this situation.[6] Trust can therefore be understood as serving emancipation—by calling for trustworthy AI, we want to guarantee that AI cannot exercise power over us in arbitrary, harmful or excessive ways.

4.8 Other Topics in AI Ethics

Although there exists a significant amount of overlap between the main debates in the field of AI ethics and the principles that are most commonly mentioned in AI ethics guidelines, these guidelines do not cover topics like machine ethics, the moral status of AI systems, or technological singularity. Moreover, while there is wide agreement on what ethical principles should be reflected in the development and use of AI, the views on these other moral questions in AI are much more varied. But, there is nevertheless a shared concern present within all of these debates. All these topics address, each in their own way, the question of how we should relate to AI and exercise control over it. AI has the potential to become an unprecedentedly powerful technology, due to its intelligence, ability to function autonomously and also widespread reliance on technology. So debates regarding the norms according to which we want AI to act, whether we should grant AI rights, and whether the technology poses an existential risk or not, all express a concern for the human's position in relation to the (potential) power of AI.

[6] For a more elaborate discussion of the relation between power and the moral concepts of trust, vulnerability, authenticity, and responsibility, see Sattarov 2019.

4.9 Non-Western Perspectives on AI Ethics

One of the criticisms raised against the principled approach to AI ethics is that the guidelines that have been established (including the ones discussed by Jobin et al., 2019) only represent Western views and values (Gordon & Nyholm, 2021). An AI ethics based on non-Western philosophy (e.g., Daoism, Confucianism, or Ubuntu) might not focus as much, or perhaps not at all, on the individual and their emancipation and empowerment. However, Gal (2020) shows that AI ethics guidelines developed in South Korea, China, and Japan show a lot of similarities with the guidelines studied by Jobin et al. (2019). The South Korean government presented an ethical framework in 2018, aimed at achieving a human-oriented intelligent information society. The framework consisted of four main principles: publicness, accountability, controllability, and transparency. Central to the Korean approach, Gal notes, is "a clear human-over-machine hierarchy" (Gal, 2020, 609). Chinese approaches show the same emphasis on the idea that AI should first and foremost be a tool to benefit humans. Furthermore, China's own big tech companies Baidu and Tencent developed AI ethics principles that show a lot of convergence with the principles mentioned before. Baidu lists safety and controllability, equal access, human development, and freedom. Tencent says AI should be available, reliable, comprehensive, and controllable. Japan deviates most from the trend, by envisioning a coexistence and coevolution between humans, on the one side, and AI and robots on the other side.

So although it is not unimaginable to have AI ethics approaches with an entirely different focus, the current state of AI ethics is that the field is predominantly concerned with human emancipation and empowerment. The goal of AI ethics is to ensure that the emerging technologies that promise to radically change life as we know it, do so for the better.

5 AI ethics as a Critical Theory

By defining the ethical principles and moral questions that are central in AI ethics in terms of power (under a pluralist understanding of power, that is), I have shown that the field is driven by a fundamental concern for human emancipation and empowerment. Transparency, privacy, freedom, and autonomy are valued because they are empowering—they grant individuals the ability to rule their own lives. Principles like trust, justice, responsibility, and non-maleficence are important because they protect individuals against the power that could be exercised by means of AI, or possibly even by AI itself. These central concerns are usually not made explicit. However, doing so helps us to see that AI ethics resembles a critical theory. Like any critical theory, the purpose of AI ethics is not merely to analyze or diagnose society, but also to change it. Both critical theory and AI ethics have a practical goal, namely that of empowering individuals and protecting them against systems of power. But while critical theory is concerned with society at large, AI ethics focuses on the part that a particular type of technology plays in society. Hence, we could say that AI

ethics is a critical theory, which focuses on the ways in which human emancipation and empowerment are or could be hindered by AI technology.

Understanding AI ethics as a critical theory can help the field forward in a number of ways. First of all, defining ethical principles in terms of their relation to emancipation or empowerment (respectively, relational, or dispositional power) creates a common language to compare ethical issues and to discuss them in interdisciplinary contexts. Such a common language is welcome for two reasons: because the principles have been accused of being too abstract (Mittelstadt, 2019; Resseguier, 2021; Ryan and Stahl, 2020) and because improving AI is a necessarily interdisciplinary endeavor. Not just ethicists, but also tech developers, data scientists, policymakers, and legal experts need to be involved to realize the goal of ethical AI. The language of power could be more appropriate in interdisciplinary contexts than discussions about (sometimes highly contested) moral values and principles are. Furthermore, by defining how ethical issues relate to dispositional and relational power, we immediately have a clear target for change. While the principles of transparency, privacy, or justice might not be action guiding as such, empowering individuals with respect to their knowledge or informational control, or reducing the say AI systems have over their lives, are much more tangible goals.

A second benefit of the insight that AI ethics is a critical theory is that it offers us a new method for identifying and analyzing ethical issues: a power analysis. The principled approach to AI ethics functions as a kind of ethical checklist. However, the potential negative implications of a technology (and to some extent also the positive ones) could also be identified by analyzing in what ways the technology limits a person's agency or freedom. The different aspects of power that I described in Sect. 3 could inform and guide such a power analysis. In addition to being more appropriate for interdisciplinary work, a power analysis also has the advantage that it could cover ethical issues that are left unaddressed by current AI principles. The principled approach has been accused of giving too little attention to the social and political context in which AI applications are developed and used, and because of which ethical issues arise. Using a power analysis to ethically assess emerging AI systems will improve our understanding of the ways in which ethical issues in AI tie into broader social, political, economic and historical matters, and understanding the broader context of an ethical issue will in turn make it easier to address the issue, not just on a technical level, but on a societal and political level.

Furthermore, a power analysis could be complemented by insights from the tradition of critical theory. Critical theory is not a full-fledged normative theory that explains what is right and what is wrong in the way that classic theories like consequentialism, deontology, or virtue ethics do, but it does take in a normative stance. Just like critical theory, AI ethics is not meant to be an ethical theory in the classic sense, but it should diagnose technological advancements in society and change them for the better. As others have argued before (Delanty & Harris, 2021; Feenberg, 1991; Fuchs, 2016), critical theory offers a valuable toolbox for analyzing the societal implications of modern technologies. I add to these arguments that many of these societal implications tie into ethical issues. Or in other words, critical theory can help to pinpoint ethically relevant issues that are not typically

addressed by ethical principles or classic ethical theories. Critical theory could, for example, help to understand ethical issues that arise from AI's relation to present-day capitalism (following first-generation critical theorists) or the potential ethical implications of misrecognition that is mediated by AI (following Honneth, 1996).

A final benefit of unmasking AI ethics as a critical theory is that we can now understand AI ethics principles as having a common aim. Mittelstadt (2019) criticizes the popular AI ethics principles for lacking a common aim. Without such a common aim, Mittelstadt argues, the principled approach to AI ethics could not have the same success as the principled approach in bioethics has. But by defining the AI ethics principles in terms of power, I showed that they do share a common aim: to protect human emancipation and empowerment in the face of this new, powerful technology. AI might imply technological progress, but that does not guarantee social progress. AI that decrease our freedom, our agency, and our ability to develop ourselves, would be a step back for the emancipation of individuals and societal groups. As Forst writes, "every progressive process must be constantly questioned as to whether it is in the social interest of those who are part of this process" (Forst, 2019, 21).

6 Conclusion

The emerging field of AI ethics is unlike other fields in applied ethics. At the center of its attention is not human conduct, but the ways in which humans are affected by AI technology. It differs from the general ethics of technology too, in the sense that AI comes with radically new possibilities for action. This not only raises new moral questions, but also requires new approaches to conduct ethical analyses. But the most popular approach thus far—that is, the principled approach—has been met with criticism. Although there is a lot of convergence when it comes to determining which ethical issues or principles should shape the development, policy, and use of AI, AI ethics principles have been accused of being too abstract, little action guiding, and insufficiently attuned to the social and political context of ethical issues.

The aim of this paper was to show that AI ethics is just like a critical theory. I have explained that a critical theory is aimed at diagnosing and changing society for emancipatory purposes. I then showed that both the big debates in AI ethics and the most common AI ethics principles are fundamentally concerned with either individual empowerment (dispositional power) or the protection of those subjected to power relations (relational power). Approaching AI ethics as a critical theory, by diagnosing AI's impact by means of a power analysis and the insights of critical theory, can help to overcome the shortcomings of the currently dominant principled approach to AI ethics. Further research could test the power analysis out in a concrete case study, further assess the extent to which the understanding of AI ethics as a critical theory resonates with non-western approaches to AI ethics, or investigate how related field (such as machine ethics) could benefit from the critical theory perspective.

Funding The author is an early-stage researcher funded by MSCA ITN PROTECT project. This project has received funding from the European Union's Horizon 2020 research and innovation program under the Marie Skłodowska-Curie grant agreement No 813497.

Data availability Not applicable.

Declarations

Ethics Approval and Consent to Participate Not applicable.

Consent for Publication Not applicable.

Competing Interests The author declares no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Allen, A. (2016). Feminist perspectives on power. *The Stanford Encyclopedia of Philosophy, edited by E.N. Zalta.* Accessed October 15, 2021. https://plato.stanford.edu/archives/fall2016/entries/feminist-power/

Allen, A. & Mendieta, E. (2019). Introduction. In *Justification and emancipation. The critical theory of Rainer Forst*. Edited by Amy Allen and Eduardo Mendieta. The Pennsylvania State University Press.

Bohman, J. (2021). Critical Theory. *The Stanford Encyclopedia of Philosophy, edited by E.N. Zalta.* Accessed October 15, 2021. https://plato.stanford.edu/cgi-bin/encyclopedia/archinfo.cgi?entry=critical-theory

Brey, P. (2008). The technological construction of social power. *Social Epistemology, 22*(1), 71–95. https://doi.org/10.1080/02691720701773551

Cave, S., & Dihal, K. (2020). The Whiteness of AI. *Philosophy & Technology, 33*(4), 685–703. https://doi.org/10.1007/s13347-020-00415-6

Cobbe, J. (2020). Algorithmic censorship by social platforms: Power and resistance. *Philosophy & Technology*. https://doi.org/10.1007/s13347-020-00429-0

Dahl, R. A. (1957). The concept of power. *Behavioral Science*, 201–215. https://doi.org/10.7312/popi17594-004

Danaher, J. (2020). Freedom in an age of algocracy. *Oxford Handbook on the Philosophy of Technology*, 1–32.

de Laat, P. B. (2019). The disciplinary power of predictive algorithms: A Foucauldian perspective. *Ethics and Information Technology, 21*(4), 319–329. https://doi.org/10.1007/s10676-019-09509-y

DeCew, J. (2018). Privacy. Zalta, E.N. (Ed.). *The Stanford Encyclopedia of Philosophy*. First edition 2018. Accessed October 15, 2021. https://plato.stanford.edu/archives/spr2018/entries/privacy/

Delanty, G., & Harris, N. (2021). Critical theory and the question of technology: The Frankfurt School revisited. *Thesis Eleven, 166*(1), 88–108.

Dubber, M.D., Pasquale, F. and Das, s. (eds) (2020). *The Oxford handbook of ethics of AI*. Oxford University Press. https://doi.org/10.1093/oxfordhb/9780190067397.001.0001

Engels, F. (1886). *Ludwig Feuerbach und der Ausgang der klssischen deutschen Philosophie*. Stuttgart: Neue Zeit.

Feenberg, A. (1991). *Critical theory of technology*. Oxford University Press.

Forst, R. (2015). Noumenal Power. *Journal of Political Philosophy, 23*(2), 111–127. https://doi.org/10.1111/jopp.12046

Forst, R. (2019). The justification of progress and the progress of justification. In *Justification and emancipation. The critical theory of Rainer Forst*. Edited by Amy Allen and Eduardo Mendieta. The Pennsylvania State University Press.

Fuchs, C. (2016). *Critical theory of communication: New readings of Lukács, Adorno, Marcuse, Honneth and Habermas in the age of the Internet.* University of Westminster Press.

Gal, D. (2020). Perspectives and approaches in AI ethics. East Asia. In *The Oxford Handbook of Ethics of AI*. Edited by M. D. Dubber, F. Pasquale, and S. Das. Oxford University Press. DOI https://doi.org/10.1093/oxfordhb/9780190067397.001.0001

Gebru, T. (2020). Race and Gender. In *The Oxford handbook of ethics of AI*. Edited by M. D. Dubber, F. Pasquale, and S. Das. Oxford University Press. DOI https://doi.org/10.1093/oxfordhb/9780190067397.001.0001

Gordon, J. S. & Nyholm, S. (2021). Ethics of artificial intelligence. *Internet Encyclopedia of Philosophy.* Accessed January 21, 2022. https://iep.utm.edu/ethic-ai/

Habermas, J. (1984). The Theory of Communicative Action. Vol. I: Reason and the Rationalization of Society. Translated by T. McCarthy. Boston: Beacon Press. [Published in German in 1981]

Habermas, J. (1987). The theory of communicative action. Vol. II: Lifeworld and System. Translated by T. McCarthy. Boston: Beacon Press. [Published in German in 1981]

Haugaard, M. (2010). Power: A "family resemblance concept." *European Journal of Cultural Studies, 13*(4), 419–438.

Haugaard, M. (2020). *The four dimensions of power: Understanding domination, empowerment and democracy*. Manchester University Press.

Honneth, A. (1996). *The struggle for recognition: The moral grammar of social conflicts*. MIT Press.

Horkheimer, M. & Adorno, T.W. (2002). *Dialectic of Enlightenment.* Translated by Edmund Jephcott, edited by Gunzelin Schimdt Noeri. Stanford University Press.

Horkheimer, M. (1972). *Critical theory selected essays*. Translated by Matthew J. O'Connell. New York: The Continuum Publishing Company.

Jobin, A., Ienca, M. & Vayena, E. (2019). The global landscape of AI ethics guidelines. *Nature Machine Intelligence*. 10.1038/ s42256-019-0088-2

Lukes, S. (1974). *Power: radical view*. London: Macmillan.

Lukes, S. (2005). *Power: A radical view. Second edition*. London: Red Globe Press.

Mittelstadt, B. (2019). Principles alone cannot guarantee ethical AI. *Nature Machine Intelligence*. https://doi.org/10.1038/s42256-019-0114-4

Mohamed, S., Png, M. T., & Isaac, W. (2020). Decolonial AI: Decolonial theory as sociotechnical foresight in artificial intelligence. *Philosophy & Technology, 33*, 659–684.

Morriss, P. (2002). *Power: A philosophical analysis*. Manchester, New York: Manchester University Press.

Müller, V. (2020). Ethics of artificial intelligence and robotics. Zalta, E.N. (Ed.). *The Stanford Encyclopedia of Philosophy*. First edition 2020. Accessed January 21, 2022. https://plato.stanford.edu/entries/ethics-ai/

Noble, S.U. (2018). *Algorithms of oppression: How search engines reinforce racism*. New York: New York University Press.

Resseguier (2021). Ethics as attention to context: Recommendations for AI ethics. In *SIENNA D5.4: Multi-stakeholder strategy and practical tools for ethical AI and robotics*. https://www.sienna-project.eu/publications/deliverable-reports/

Ryan, M., & Stahl, B. C. (2020). Artificial intelligence ethics guidelines for developers and users: Clarifying their content and normative implications. *Journal of Information, Communication and Ethics in Society, 19*(1), 61–86. https://doi.org/10.1108/JICES-12-2019-0138

Sattarov, F. (2019). *Power and technology: A philosophical and ethical analysis*. London, New York: Rowman and Littlefield International.

Stahl, B. C., Doherty, N. F., Shaw, M., & Janicke, H. (2014). Critical theory as an approach to the ethics of information security. *Science and Engineering Ethics, 20*, 675–699. https://doi.org/10.1007/s11948-013-9496-6

Thompson, S. (2006). *The political theory of recognition. A critical introduction*. Polity Press.

Zuboff, S. (2019). *The age of surveillance capitalism: The fight for a human future at the new frontier of power*. London: Profile Books.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.